PROJECT PROPOSAL

Subject: Programming For Bioinformatics (SECB3202)

Session: 2025/2026 Semester 1

Title: Pancreatic Cancer Prediction

Group Name	Group 10
Group Members	1. RAVINESH A/L MARAN (A23CS0175) 2. WELSON WOONG LU BIN (A23CS0196) 3. BERNICE LOU MIN YUN (A23CS0056)
Section	01
Lecturer's Name	DR. SEAH CHOON SEN

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1.0 Introduction

Pancreatic cancer is one of the most lethal human cancers; it is characterized by late diagnosis and rapid disease progression, leading to very poor survival rates. Symptoms are often subtle or mistaken for common gastrointestinal conditions that make early detection difficult. The global statistics on cancer indicate that the 5-year survival rate for pancreatic cancer is less than 10%, mainly due to the absence of reliable methods for its screening and biomarkers.

Recently, bioinformatics and machine learning have become key tools to predict cancer risk and identify the most important clinical indicators. The analysis of patient medical, clinical, and lifestyle data by machine learning models can reveal hidden relationships that may improve early detection and risk assessment of cancer.

This work is designed to apply a bioinformatics approach in the prediction of pancreatic cancer progression, analysis of key biomarkers, and investigation into how demographic, clinical, and lifestyle variables may influence patient outcomes.

1.1 Problem Background

Pancreatic cancer usually develops silently, and it is very hard to clearly identify early symptoms. The risk factors-smoking habits, obesity, diabetes, hereditary history, and chronic pancreatitis-all contribute to the progression of cancer. These factors are all interconnected, so doctors may not be able to interpret each and every one of them. Large datasets including symptoms of patients, tumor stages, lifestyle habits, types of treatment, socioeconomic conditions, physical activity level, and alcohol consumption allow for deeper analysis. Using bioinformatics approaches, these patterns can be more systematically studied to help predict cancer stage and identify high-risk individuals.

1.2 Problem Statement

Although clinical data on pancreatic cancer patients is indeed available, there still lacks a bioinformatics model that can automatically handle several parameters concerning a patient in order to predict the severity of cancer. The health expert might face difficulties in interpreting such high-dimensional data and understanding which factors most contribute to disease conditions. This leads to poor early detection, and as a result, patients usually get diagnosed when the disease has progressed to a higher stage. This calls for the development of a machine-learning-based bioinformatics model which can predict cancer stage or survival likelihood and identify the most influential risk factors.

1.3 Objectives

The main objective of this project is to develop a simple machine-learning prediction model for pancreatic cancer using clinical and lifestyle data. This project aims to preprocess and explore the dataset to identify important patterns, apply suitable classification methods to predict cancer stage or outcomes, and determine which factors act as key biomarkers. Another objective is to evaluate the performance of the model using common metrics such as accuracy or confusion matrix so that the reliability of the prediction can be understood.

1.4 Scopes

This project will focus on the Pancreatic Cancer Prediction Dataset from Kaggle. The scope includes cleaning the data, handling missing values, encoding categorical data, and performing simple exploratory analysis. The project will then apply selected machine learning classification techniques using Python libraries such as Pandas, NumPy, and Scikit-Learn. Only computational analysis will be performed; no real clinical diagnosis or laboratory experiments are included. The project also does not involve genomic or deep-learning-based approaches. All progress will be documented on GitHub as required by the course.

1.5 Conclusion

In summary, this project aims to use bioinformatics and machine-learning techniques to study clinical and lifestyle factors related to pancreatic cancer. By analysing the dataset, the project hopes to identify patterns, predict cancer severity, and highlight important risk factors involved in disease progression. This work supports the course requirement for applying programming and data analysis skills to a biological problem while demonstrating how computational tools can help improve understanding of complex diseases.